

AIQG — Agent Identity & Per-Flow Attribution: Prior-Art Research

Reference findings that informed [AIQG-AGENT-FLOW-ATTRIBUTION.md](#).

- **Date:** 2026-06-09/10
- **Method:** multi-source deep-research — 5 search angles, 22 sources fetched, 105 claims extracted, top 25 verified via 3-vote adversarial checking (need 2/3 to refute). **21 confirmed, 4 killed.** 104 agents.
- **Question:** How do systems identify an AI agent and attribute LLM/API gateway traffic *per agent flow* — grouping many calls into the agent, the run/execution, the conversation/session, and the step?

Confidence note: confirmed claims rest on primary sources (the specs and product docs themselves), so confidence reflects faithful reading of authoritative documentation, not independent cross-vendor benchmarking.

Headline

A consistent **four-level layering** emerges everywhere — *agent* → *conversation(session/thread)* → *run/flow* → *step* — but the **trust model splits**: production observability tooling attributes traffic via **self-asserted** ids the caller picks (headers/metadata), while emerging agent-identity protocols make identity **cryptographically authenticated**. That split is the central design choice for an LLM gateway: descriptive (easy, today) vs. trustworthy (frontier, billing-grade).

Confirmed findings by area

Standards — OpenTelemetry GenAI semantic conventions (the canonical scheme)

Status: **experimental** / “**Development**” — names may change.

- **Agent:** `gen_ai.agent.id` (“unique identifier of the GenAI agent”, e.g. `asst_5j66...`) + `gen_ai.agent.name` (“human-readable name ... provided by the application”, e.g. `Math Tutor`) + `gen_ai.agent.description` + `gen_ai.agent.version`.
- **Conversation/session/thread:** a **single** `gen_ai.conversation.id` — “The unique identifier for a conversation (session, thread)...” — *folds session and thread into one id*. There is **no** `gen_ai.session.id` or `gen_ai.thread.id`. (A generic cross-domain `session.id` exists outside the `gen_ai` namespace, not cross-referenced.)
- **Step:** an OTel **span**; agent operations are named via `gen_ai.operation.name` ∈ {`create_agent`, `invoke_agent`, `invoke_workflow`, `execute_tool`, `chat`, `embeddings`, `retrieval`, ...}, with span names like `invoke_agent {gen_ai.agent.name}`.
- **Run/flow & correlation:** ordinary trace/span parent-child structure (`trace_id` / `span_id` / `parent_span_id`).
- Sources: `opentelemetry.io/gen-ai/gen-ai-agent-spans/`, `/registry/attributes/gen-ai/`, `/gen-ai/gen-ai-spans/`.

Industry tools — same layering, different names, mostly self-asserted

- **LangSmith (LangChain):** run-tree model — `run_id` = the span/step, `trace_id` = the root trace/run (equals the root run’s own id), `parent_run_id` = tree linkage (`dotted_order` encodes hierarchy, `RunTree` auto-links via context). Session layer = **any one of three interchangeable** metadata keys: `session_id` / `conversation_id` / `thread_id`. Docs: docs.langchain.com/langsmith (export-traces, run-data-format).

- **Langfuse:** OTEL-aligned ids — **32-hex trace ids, 16-hex observation ids** (W3C 16-byte/8-byte sizing). Session = free-form sessionId (US-ASCII, <200 chars, app-chosen; 1:n to traces; groups all observations - enclosing traces). Propagation: **SDK-managed context** (propagate_attributes), with an **opt-in as_baggage** flag to propagate via W3C Baggage HTTP headers cross-service. Docs: langfuse.com/docs/observability.
- **Helicone (the pure-header gateway model):** three **client-supplied HTTP headers** — Helicone-Session-Id (UUID recommended; groups all related requests), Helicone-Session-Path (STEP hierarchy via / parent-child tree, e.g. /abstract/outline/lesson-1), Helicone-Session-Name. **Any** request type (LLM, vector DB, tool call) sharing the same session headers is folded into one session trace. Purely value-based and self-asserted. Docs: docs.helicone.ai/features/sessions.

Propagation mechanism

- Libraries: **SDK-managed trace context** (OTel context inheritance, Langfuse propagate_attributes, LangSmith RunTree).
- Gateways: **HTTP headers** (Helicone), optionally honoring **W3C traceparent / Baggage**. A gateway cannot ride the caller's SDK context, so headers are the practical mechanism.

Agent-identity protocols — the authenticated frontier (diverges)

- **AIP (Agent Identity Protocol)** — two id schemes: DNS-based aip:web:<domain>/<path> (resolved at https://<domain>/.well-known/aip/<path>.json) and self-certifying aip:key:ed25519:<multibase> (the id is the public key). Identity documents **MUST** carry a document_signature (Ed25519 over RFC 8785 canonical form) — **“not self-asserted — it is cryptographically authenticated”**, tamper-evident even if the hosting domain is compromised. Correlation across protocols uses **Invocation-Bound Capability Tokens (IBCTs)** with transport bindings: X-AIP-Token HTTP header (MCP tool calls), aip_token task-metadata field (A2A), Authorization: AIP <token> (plain HTTP). A credential-bound correlation artifact, not a self-asserted id. Sources: arXiv 2603.24775, IETF draft-prakash-aip-00. Status: 2026 preprint + early I-Ds, not ratified; adoption unproven.
- Related (lower-confidence / blog): Google **A2A** protocol spec, **WSO2 AI Gateway “AgentID”** (gateway-level agent identity/auth).

Killed claims (failed verification — do NOT rely on)

- ☐ “gen_ai.operation.name is a *required* attribute” (0-3) — it is conditionally required, not required.
- ☐ “Langfuse uses a four-part trace/observation/span/parent model” (0-3) — it is trace + observation (16-hex), not a separate span/parent split.
- ☐ “LangSmith grouping ids are metadata-only with no propagation mechanism” (0-3) — they propagate via the RunTree/context model.
- ☐ “GenAI correlation relies on no GenAI-specific propagation field; the spans page defines no propagation” (1-2, not killed but contested).

Coverage gaps (absence of evidence ≠ evidence of absence)

- **Academic papers:** no paper-level correlation model survived verification — the scholarly landscape here is **unconfirmed**.
- **Other tools:** Arize Phoenix, Traceloop/OpenLLMetry, Datadog LLM Observability, Honeycomb, W&B Weave produced **no confirmed claims**.

- **W3C Trace Context/Baggage** appears only indirectly (via Langfuse sizing), not surveyed as a standalone propagation standard.
- **Patents: zero confirmed claims.** Four USPTO patents on gateway attribution/metering were *located* — **US 12,483,411 / 12,547,677 / 10,924,326 / 11,785,115** — but their claims/assignees were not verified within budget. **Freedom-to-operate is unresolved** and owed before any billing use (see task: Patent FTO pass).

Design implications (carried into the plan)

1. **Adopt OTEL gen_ai.* names** rather than inventing TAS-specific field names — standards-aligned, OTEL-export-ready. Pin names (experimental).
2. **Fold session+thread into one conversation.id** (matches OTEL + LangSmith’s interchangeable keys).
3. **Header-based propagation** for the gateway (Helicone-proven), honoring W3C trace-parent so `flow_id = trace_id`, `step = span_id` when present.
4. **Ship self-asserted, design for authenticated.** Record `identity_source` ∈ {header, trace, inferred}; reserve `agent_identity_verified` + credential ref so AIP/IBCT, token-bound, or AgentID can slot in later without migration.
5. **Patent FTO** gates billing-grade per-agent metering.

Source index (confirmed-claim sources)

- opentelemetry.io/docs/specs/semconv/gen-ai/gen-ai-agent-spans/
- opentelemetry.io/docs/specs/semconv/registry/attributes/gen-ai/
- opentelemetry.io/docs/specs/semconv/gen-ai/gen-ai-spans/
- opentelemetry.io/docs/concepts/context-propagation/
- docs.langchain.com/langsmith/export-traces/, [run-data-format](https://docs.langchain.com/langsmith/run-data-format/)
- langfuse.com/docs/observability (trace-ids-and-distributed-tracing, sessions, data-model, sdk/instrumentation)
- docs.helicone.ai/features/sessions
- arxiv.org/html/2603.24775 + IETF draft-prakash-aip-00 (AIP)
- a2a-protocol.org/latest/specification/ (A2A)
- w3.org/TR/trace-context/ (W3C Trace Context)
- USPTO patents (located, claims unverified): 12483411, 12547677, 10924326, 11785115